

Graph Sketching

Problem-1: Sketch the graph of the following function:

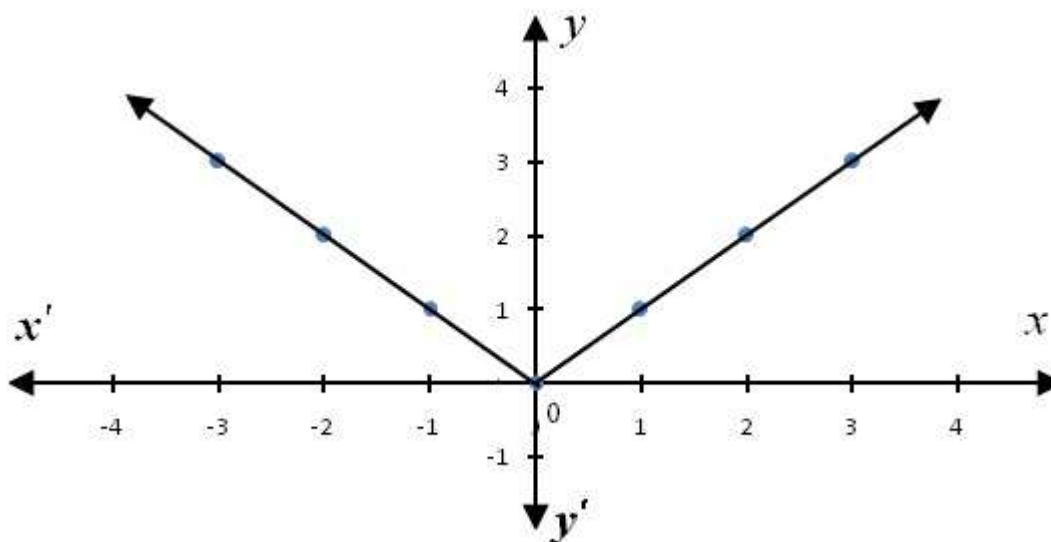
$$f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases}$$

Solution: Let $y = f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases}$

Now to draw the graph of the given function we make a table as follows:

x	3	2	1	0	-1	-2	-3
y	3	2	1	0	1	2	3

Now using the values of x and y of the above table we draw the following graph:



Problem-2: Sketch the graph of the following function:

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$$

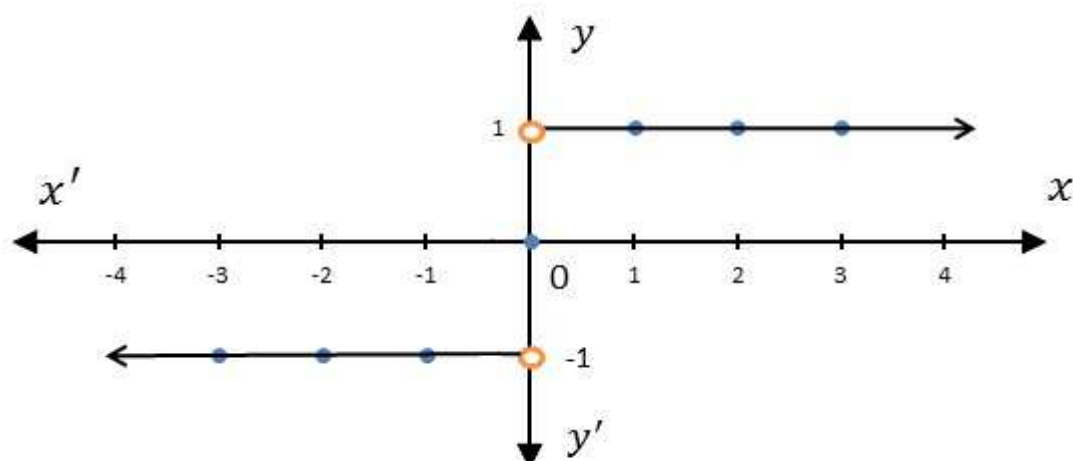
Solution:

$$\text{Let } y = f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$$

Now to draw the graph of the given function we make a table as follows:

x	3	2	1	0	-1	-2	-3
y	1	1	1	0	-1	-1	-1

Now using the values of x and y of the above table we draw the following graph:



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H.W: Problem-3: Sketch the graph of the function: $f(x) = x^2$

Problem-4: Sketch the graph of the following function:

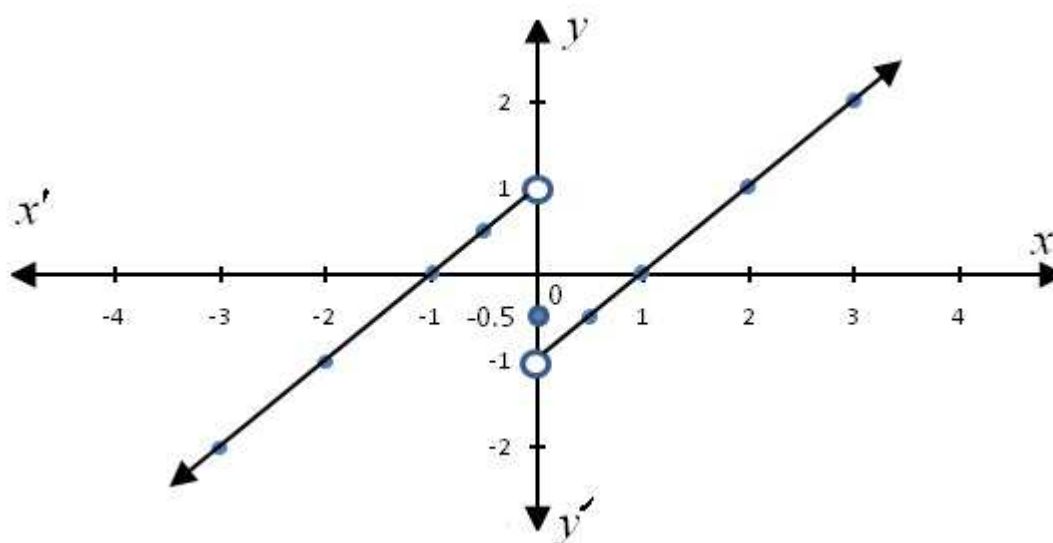
$$f(x) = \begin{cases} x - 1 & \text{when } x > 0 \\ -\frac{1}{2} & \text{when } x = 0 \\ x + 1 & \text{when } x < 0 \end{cases}$$

Solution: Let $y = f(x) = \begin{cases} x - 1 & \text{when } x > 0 \\ -\frac{1}{2} & \text{when } x = 0 \\ x + 1 & \text{when } x < 0 \end{cases}$

Now to draw the graph of the given function we make a table as follows:

x	3	2	1	.5	0	-.5	-1	-2	-3
y	2	1	0	-.5	-.5	.5	0	-1	-2

Now using the values of x and y of the above table we draw the following graph:



Problem-5: Sketch the graph of the following function:

$$f(x) = \begin{cases} -2x + 1 & \text{when } x < 0 \\ 1 & \text{when } 0 \leq x < 1 \\ 2x - 1 & \text{when } x \geq 1 \end{cases}$$

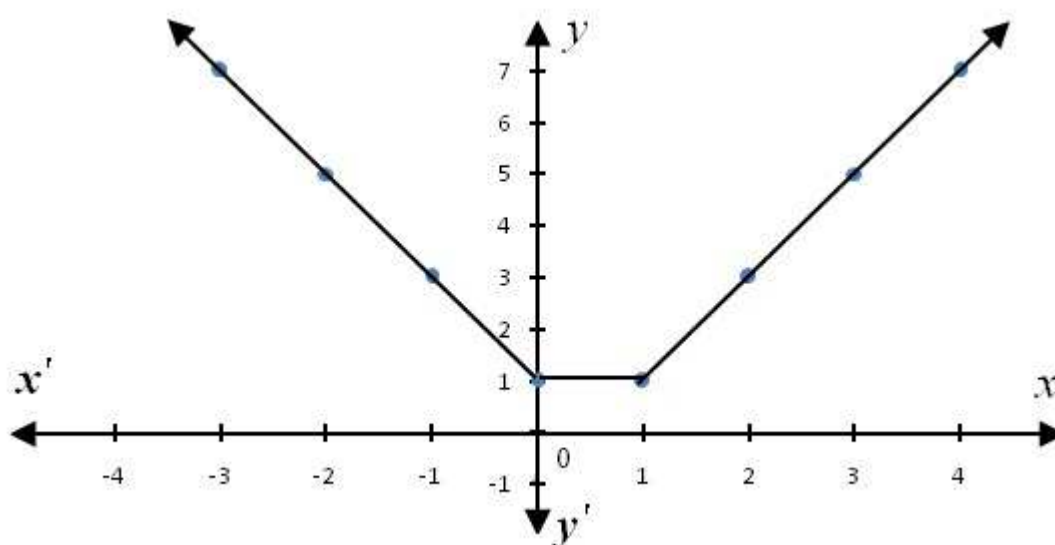
Solution: Let $y = f(x) = \begin{cases} -2x + 1 & \text{when } x < 0 \\ 1 & \text{when } 0 \leq x < 1 \\ 2x - 1 & \text{when } x \geq 1 \end{cases}$

Now to draw the graph of the given function we make a table as follows:

x	4	3	2	1	0	-1	-2	-3
y	7	5	3	1	1	3	5	7

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Now using the values of x and y of the above table we draw the following graph:



Problem-6: Sketch the graph of the following function:

$$f(x) = \begin{cases} x & \text{when } 0 \leq x < \frac{1}{2} \\ 1 & \text{when } x = \frac{1}{2} \\ 1 - x & \text{when } \frac{1}{2} < x \leq 1 \end{cases}$$

Solution: Let $y = f(x) = \begin{cases} x & \text{when } 0 \leq x < \frac{1}{2} \\ 1 & \text{when } x = \frac{1}{2} \\ 1 - x & \text{when } \frac{1}{2} < x \leq 1 \end{cases}$

Now to draw the graph of the given function we make a table as follows:

x	0	.2	.4	.5	.6	.8	1
y	0	.2	.4	1	.4	.2	0

Now using the values of x and y of the above table we draw the following graph:

